Engineering Electromagnetics

griffith's Antenna's.

5pm ↓ fri

- Transmission lines.
- Plane waves
- Plane waves at media interfaces
- Antennas (griffiths)
- wave propagation in bounded media

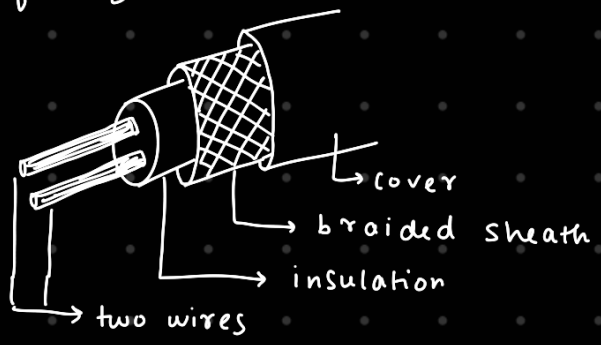
you can design a TL to work as a cap/ ind.

2 Quizzes : 25% + 25% E.S : 50%

Transmission Lines

→ They carry power (AC → 50Hz) => not a signal since it doesn't carry any information.

→ eg of signal Transmission line : Optical fibre



→ eg: mobile antennae => Flat wiggly lines
Crystal Oscillator → generates transmission frequencies.

→ eg: nose of an aeroplane: has an antenna, has radomes to protect the antenna that transmits signal.

radomes → radio frequency ← RF cables have plastic radome

↓

used in microwave communication links → to protect the antennas from weather effects.

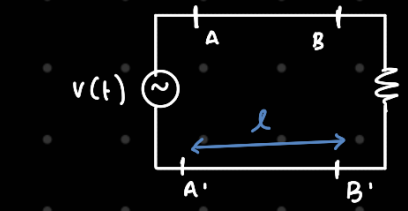
<31·7·24>

1. When to use TL theory?

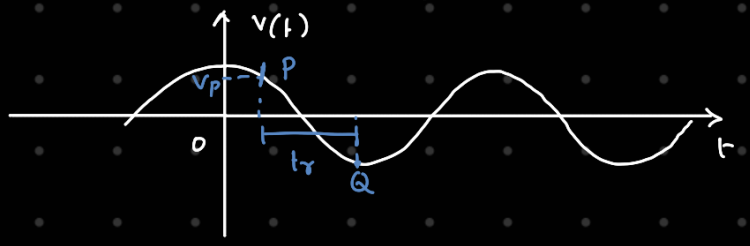
why does KCL, KVL fail? → t_r << Time period.

f

$2\pi f = \omega$



l = length of the TL.
V = velocity w/ w/I the current is flowing.
t_r = transit time = l/v



if transit time is much less compared to T, both AA' and BB' will have almost the same voltage at an instant.

↓ we can then use KCL & KVL

↓ Lumped circuit.

i.e., all points in the TL are in ϕ

eg: 50Hz TL

$T = 20\,ms = \left(\frac{1}{50Hz}\right)$

l = B/w 2 transformers = 4 km (say)

$V = 2 \times 10^8$ m/s ('c' or just less than)

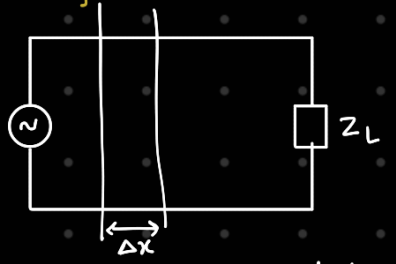
$t_r = \frac{4 \times 10^3}{2 \times 10^8} = 2 \times 10^{-5}\,s = 0.02\,ms.$ → it's 1000 times less than T

↓

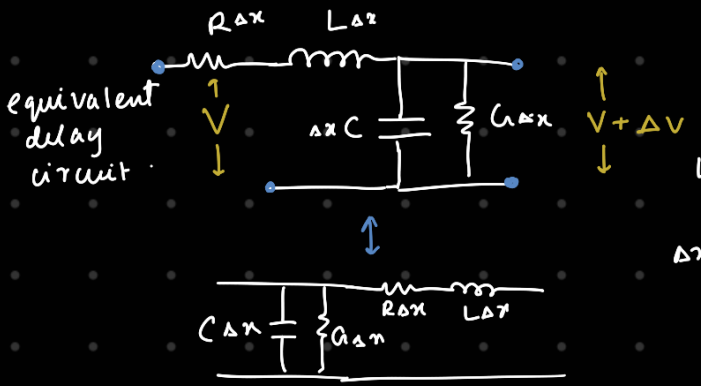
∴ electric cables are considered to be lumped circuit.

- for different V, there is different ϕ, w/I makes it difficult to transmit this signal.

2. How to analyse Transmission lines w/ circuit theory?



At low V, X_L should be 0?



ideal Capacitor has $R = \infty$

L, C, R, G /m.

Δx - infinitesimally small section

↓ for this we can apply circuit theory.

Carrier → 2.5GHz
bandwidth → 5MHz

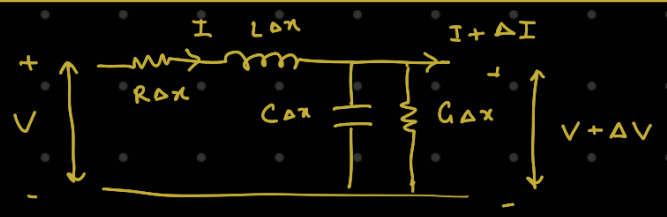
Telegrapher's =n by Oliver Heaviside

↑

Kirchhoff's corollary to multiple frequency.

↓

unit step func^n is a heaviside func^n



H.W

$\Delta V =$

$\Delta I =$

Apply Kirchhoff's line.

↓

Δx is small

↓

t_r is small

↓

KCL & KVL ✓

$$V - IR\Delta x - L\Delta x \frac{dI}{dt} - (V + \Delta V) = 0$$

only this let's us extend it to all frequencies.

↓

o/w for some specific high v, the t_r will become significant.

$$\Delta V = -IR\Delta x - L\Delta x \frac{dI}{dt} = -IR\Delta x - Lj\omega I\Delta x$$

$$\cancel{I} = C\Delta x \frac{d(V)}{dt} + G\Delta x V + \cancel{I} + \Delta I$$

$$\Delta I = -C\Delta x \frac{dV}{dt} - G\Delta x V = -C\Delta x j\omega V - G\Delta x V$$

$I = I_0 e^{j\omega t}$ $\quad V = V_0 e^{j\omega t}$ ⟶ phasor

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<1·8·24>

- L & C → for creating $\Delta\phi$
  R → energy losses.
- why is it called distributed elements?
  for every element, circuit theory is valid.

$$\frac{\Delta V}{\Delta x} = -(R + j\omega L) I$$

$$\frac{\Delta I}{\Delta x} = -(G + j\omega C) V$$

$\Delta x \to 0$

$$\frac{dV}{dx} = -(R + j\omega L) I$$

$$\frac{dI}{dx} = -(G + j\omega C) V$$

coupled differential equations.

$$\frac{d^2V}{dx^2} = -(R + j\omega L)\frac{dI}{dx} = (R + j\omega L)(G + j\omega C) V$$

$$\frac{d^2I}{dx^2} = -(G + j\omega C)\frac{dV}{dx} = (R + j\omega L)(G + j\omega C) I$$

Let $\quad (R + j\omega L)(G + j\omega C) = \gamma^2 \quad ; \quad \gamma = \alpha + j\beta$

$$\frac{d^2V}{dx^2} = \gamma^2 V$$

$$\frac{d^2I}{dx^2} = \gamma^2 I$$

} homogenous second-order derivative.

Let $V = e^{rx}$

$r = \pm\gamma$

$V_s e^{j\omega t} = V_s \cos(\omega t) + j V_s \sin(\omega t)$

Soln:

$V = V^- e^{\gamma x} + V^+ e^{-\gamma x}$; V^+, V^- } complex constants.

$I = I^- e^{\gamma x} + I^+ e^{-\gamma x}$; I^+, I^- }



Time harmonic nature is implicit.

$V(t,x) = \boxed{V^- e^{\gamma x} e^{j\omega t}} + V^+ e^{-\gamma x} e^{j\omega t}$.

$I(t,x) = \underbrace{I^- e^{\gamma x}}_{\text{phasor}} e^{j\omega t} + I^+ e^{-\gamma x} e^{j\omega t}$. } Spatio-temporal.

↓

to find V^+, V^-, you need 2 boundary conditions, values at $x=0$ & $x=l$.

$V_s e^{j\omega t} = V_s \cos(\omega t) + j V_s \sin(\omega t)$

first term:

$V^- e^{(\alpha+j\beta)x} e^{j\omega t}$

$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$

- ideal inductive reactance has to be high/low?
  ↓
  that depends on frequency.
  ↓

H.W

Problem 2: $R = 0.2\ \Omega/m$

$G = 0$

$L = 0.3\ \mu H/m$

$C = 15\ pF/m$.

H.W: Complex no. calculator.

$f = 900\ MHz \rightarrow 2G$ frequencies.

$2.5\ GHz \rightarrow 3G$ frequencies.

$26\ GHz \rightarrow 5G$

$110\ GHz \rightarrow 6G$.

find α and β

$\alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$

(a) $\alpha + j\beta = \sqrt{-\omega^2 LC + jR\omega C}$

$\alpha^2 + 2\alpha\beta j - \beta^2 = Rj\omega C - \omega^2 LC$

$2\alpha\beta = RC\omega$

$\alpha^2 - \beta^2 = -\omega^2 LC$

$\underbrace{V^+ e^{-\alpha x} e^{j(\omega t - \beta x)}}_{\text{first term}} + \underbrace{V^- e^{\alpha x} e^{j(\omega t + \beta x)}}_{\text{second term}}$.

first term ↓ forward propagation

second term ↓ backward propagation

↓

plot the real part of $V^+ e^{-\alpha x} e^{j(\omega t - \beta x)}$

if phase of V^+ is ϕ

$|V^+| e^{-\alpha x} \cos(\omega t - \beta x + \phi) = \text{Re}\{\text{first term}\}$

how much $\angle$ moved in unit time ↑

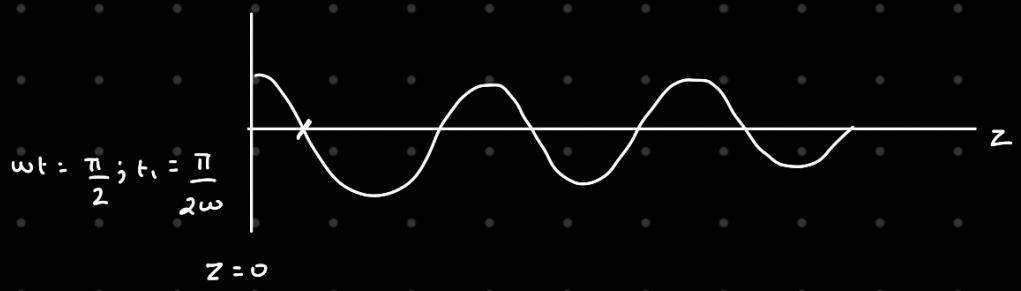
$f = \frac{\omega}{2\pi}$ (rad/s) how often wave oscillates in time ← temporal frequency.

$\beta = \frac{2\pi}{\lambda}$ ↓ attenuation in space

$\frac{1}{\lambda}$ → frequency in space.

Spatial frequency ↓ how often the wave oscillates in space (rad/m) ↓ how much angle moved in unit distance.

$t = 0$, $\phi = 0$



$\omega t = \frac{\pi}{2}$; $t_1 = \frac{\pi}{2\omega}$

$z = 0$

$\cos(\omega t - \beta x)$

$x =$		0	$\pi/2\beta$	π/β	$3\pi/2\beta$	$2\pi/\beta$
$t = 0$	$\cos\beta x$	1	0	−1	0	1
$t = \frac{\pi}{2\omega}$	$\cos(\frac{\pi}{2} - \beta x)$	0	1	0	−1	0
$t = \frac{\pi}{\omega}$	$\cos(\pi - \beta x)$	−1	0	1	0	−1
$t = \frac{3\pi}{2\omega}$	$\cos(\frac{3\pi}{2} - \beta x)$	0	−1	0	1	0

The wave is moving forward.

$d = \frac{\pi}{2\beta}$

$t = \frac{\pi}{2\omega}$

$V_p = \frac{\omega}{\beta}$ → phase velocity.

It wouldn't remain 1, it would get attenuated.

(α) ↓ exponentially decreases the amplitude as it travels along the TL.

explains signal loss.

what if?

$\frac{\omega}{\beta}$ depends on ω ↓ this leads to dispersion

β depends on ω (might not be a linear relationship)

let $\beta = k\omega^2$

$V_p = \frac{1}{k\omega}$

different frequency (components) arrive at different times → because of different speeds thru a TL.

(due to different V_p)

because $\beta = f(\omega)$

Problem 2 find ω/β

a) $f = 900\,MHz$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \sqrt{jRG\omega C - \omega^2 LC}$$

$$\gamma = \sqrt{j\ 0.2 \times 2\pi \times 9 \times 10^8 \times 15 \times 10^{-12} - (2\pi)^2 (9 \times 10^8)^2\, 0.3 \times 10^{-6}}$$

$$= \sqrt{169.56 \times 10^{-4}\, j - 958.35312 \times 10^{13}}$$

$\alpha = 6 \times 10^{-6}$

$\beta = 478.95511 \times 10^{-5}$

b) 2.5 GHz

$$\gamma = \sqrt{j \times 0.2 \times 2\pi \times 2.5 \times 10^{9} \times 15 \times 10^{-12} - (2\pi)^2 (2.5 \times 10^{9})^2 \times 0.3 \times 10^{-6} \times 15 \times 10^{-12}}$$

$$= \sqrt{j\, 18.84 \times 10^{-12} f - 11.83152 \times 10^{-3} f^2}$$

$$= \sqrt{4.7 \times 10^{-2} - 1.1 \times 10^{3}}$$

$\alpha =$ $\beta =$

c) $f = 26$ GHz

$$\gamma = \sqrt{1.9 \times 10^{-11} \times f - 1.8 \times 10^{-16} f^2}$$

$\alpha = 5.476 \times 10^{-7} \cdot s^{-1}$

$\beta = 8.94321 \times 10^{-8}$

d) $f = 110$ GHz

$$\gamma = \sqrt{2072.4 \times 10^{-3} j - 143161.392 \times 10^{15}}$$

$\alpha = 0.000000231$

$\beta = 378.3667 \times 10^{-7}$

---

what's the reason for backward propagation ? reflection.

< 2.8.24 >

$Z \equiv$ Refractive index

Forward wave

Backward wave

how much power on load & how it gets affected.

Problem: $f = 100$ GHz (6G) length of the line for which circuit theory ~~Tx line theory~~ holds good?

for dx part of Tx line

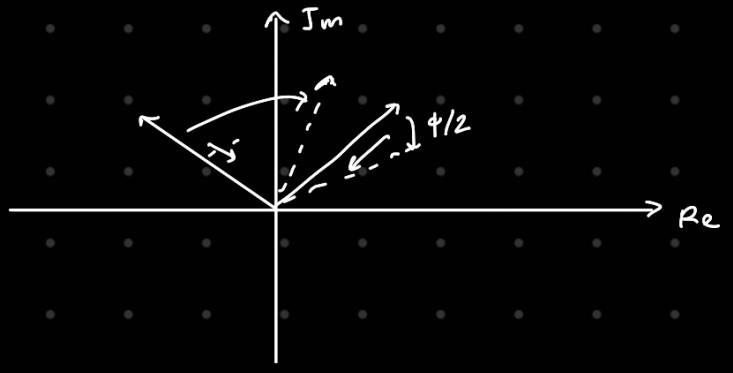
$$\lambda = \frac{c}{v} = \frac{2 \times 10^{8}}{10^{2} \times 10^{9}} = 2 \times 10^{-3}\ m$$

$l << 2 \times 10^{-3}\ m$

$l << 2\ mm$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta$$

$$= \sqrt{RG - \omega^2 LC + j\omega(LG + RC)} = \alpha + j\beta$$



(always) $Im > 0$

$\therefore \alpha, \beta > 0$.

when you square root, $\theta \rightarrow \theta/2$. → so always ends up in Q1.

- Irrespective of $\omega, R, L, G, C \rightarrow \alpha, \beta$ are positive.

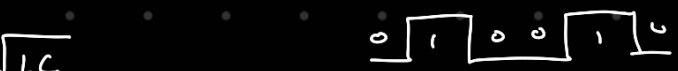
LOSS-LESS LINES:

$R = 0$

$G = 0$

$\gamma = j\omega\sqrt{LC} \Rightarrow \alpha = 0, \beta = \omega\sqrt{LC}$

$\frac{\omega}{\beta} = \frac{1}{\sqrt{LC}}$ = speed of light.



ω = sinc func^n

↓

if low pass filter the sinc

↓

you get a sinusoid.

↓

more sharp the edges

↓

more the frequencies.

↓

the wave doesn't reach the same way it started.

DISTORTIONLESS LINES: → (Vp const., $\alpha \neq f(\omega)$) no distortn in its shape as it travels.

$\frac{\omega}{\beta}$ = constant.

$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)} = \alpha + j\beta$

$\gamma = \sqrt{LC\left(\frac{R}{L} + j\omega\right)\left(\frac{G}{C} + j\omega\right)}$

Demand $\frac{R}{L} = \frac{G}{C}$

↓

β becomes a linear function of ω and α becomes frequency independent.

↓

ensures all frequency components of the signal travel at the same velocity and w/ the same attenuation,

↓

thereby, preventing dispersion and preserving the signal shape

$\gamma = \sqrt{LC\left(\frac{R}{C} + j\omega\right)^2} = \sqrt{LC}\left(\frac{R}{L} + j\omega\right)$

$\beta = \omega\sqrt{LC}$

$(R \& G \approx 0)$

$v = \frac{\omega}{\beta} = \frac{\omega}{\omega\sqrt{LC}} = \frac{1}{\sqrt{LC}}$ ← regardless of ω, same velocity & same α.

$\text{Sinc}(t) = \frac{\sin(\pi t)}{\pi t}$ → FT is a rectangular func^n → a perfect band limited signal.

$FT\{\text{sinc}(t)\} = \text{rect}(f)$

$$\text{rect}(f) = \begin{cases} 1 & , \ |f| \le \frac{1}{2} \\ 0 & , \ 0/w \end{cases}$$

when you low pass-filter a signal to an extreme degree (very low cutoff frequency), you increasingly approximate a sinusoid b/c the higher harmonics and frequency components are removed.

- If the length of Tx line is $\sim \frac{\lambda}{10}$, greater than that will require Tx line theory.

finite difference method.

- Distributed element model –
  - ↳ Telegrapher's $=^n$

telegrapher's $=^n$.

$$\frac{d^2V}{dx^2} = \gamma^2 V$$

$$\frac{d^2I}{dx^2} = \gamma^2 I$$

(A) —
$$V = V^+ e^{-\gamma x} + V^- e^{\gamma x}$$
$$I = I^+ e^{-\gamma x} + I^- e^{\gamma x}$$
($e^{-\gamma x}$ — fwd propagating; $e^{\gamma x}$ — backward propagan.)

$$\gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}$$

R, L, G, C → primary constants.

↓

we need to find V^+, V^-, I^+, I^-

it is understood that $e^{j\omega t}$ is part of the phasor representa^n ↑

$e^{j\omega t}$ is implicit here ?

↓

$$V(z,t) = \text{Re}\{V(z)\, e^{j\omega t}\}$$

($e^{j\omega t}$ ↓ time-harmonic varia^n $e^{j\omega t}$)

↓

we assume all time-varying signals can be expressed as phasors multiplied by $e^{j\omega t}$.

↓

this simplifies PDEs into ODEs.

$$\frac{dV}{dx} = -(R + j\omega L) I \quad \text{——} \ (1)$$

Substitute the solution (A) to $=^n$ (1)

$$-V^+ \gamma e^{-\gamma x} + V^- \gamma e^{\gamma x} = -(R + j\omega L)(I^+ e^{-\gamma x} + I^- e^{\gamma x})$$

$$V^+ = \sqrt{\frac{(R+j\omega L)}{(G+j\omega C)}}\, I^+$$
$$V^- = -\sqrt{\frac{(R+j\omega L)}{(G+j\omega C)}}\, I^-$$

$\Leftarrow$ $e^{-\gamma x}[\underbrace{-\gamma V^+ + (R+j\omega L) I^+}_{0}] - e^{-\gamma x}[\underbrace{-(R+j\omega L) I^- - \gamma V^-}_{0}]$ $= 0$

$$\frac{V^+}{I^+} = \left(\frac{\gamma}{R+j\omega L}\right)^{-1} = \left(\frac{\sqrt{(R+j\omega L)(G+j\omega C)}}{(R+j\omega L)}\right)^{-1}$$

$$Z_0 = \frac{V^+}{I^+} = \sqrt{\frac{(R+j\omega L)}{(G+j\omega C)}} \quad ; \quad \frac{V^-}{I^-} = -\sqrt{\frac{(R+j\omega L)}{(G+j\omega C)}}$$

Z_0 ↓ characteristic impedance

$\frac{V^+}{I^+}$ ↕ this impedance is not of the source / load.

$\sqrt{(R+j\omega L)}$ ↓ isn't this R, L, G, C just for a Δx part?

↳ these R, L, G, C are considered to be uniformly distributed along the length of the line. ↗ Constant for a homogeneous Tx line.

$\frac{V^-}{I^-} = -$ just means that wave is propagating backwards.

- Secondary parameters

$\alpha, \beta, (\gamma), Z_0$

$$V = V^+ e^{-\gamma x} + V^- e^{\gamma x}$$

$$I = \frac{V^+}{Z_0} e^{-\gamma x} - \frac{V^-}{Z_0} e^{\gamma x}$$

for reasons like ←
1. minimizing reflecⁿ
2. efficient power transfer.
3. signal integrity.

for ↑ v cables it is necessary to know the impedance.

2 boundary condⁿ at load, source.



remote radio.
↑
Base Stⁿ

where you keep the load (reference)

$x = -l$

$$V = V^+ e^{\gamma l} + V^- e^{-\gamma l}$$

$$I = \frac{V^+}{Z_0} e^{\gamma l} - \frac{V^-}{Z_0} e^{-\gamma l}$$

fallstad. transm.

Time domain Reflectometry.
↓ used to locate faults in Tx lines.

analyse the reflected signal
⇑
sending a short-duran pulse →

→ for a unit pulse:
It appears first, the disappears. → you see the unit pulse in twice the transit time.

→ these can be inferred { α, V_p } from looking at the signal from the load

→ Actual impedance, is different at different points B/c of different total V & I at different points.

- REFLECTION COEFFICIENT:

$$\Gamma(l) = \frac{V^- e^{-\gamma l}}{V^+ e^{\gamma l}} = \frac{V^-}{V^+} e^{-2\gamma l}$$

$y(x,t) = 2A\cos(kx)\cos(\omega t)$
↓
$y_1 = A\cos(kx - \omega t)$
+
$y_2 = A\cos(kx + \omega t)$

check for =ⁿ of standing wave.
↓
v have to be the same
↓
Amp can be different

$$\Gamma_L = \Gamma(l=0) = \frac{V^-}{V^+}$$

$$\boxed{\Gamma(l) = \Gamma_L e^{-2\gamma l}}$$

you might not get 0 Amp supermode.

Total wave: Standing wave

$$V = V^+ e^{\gamma l} + V^- e^{-\gamma l}$$
$$= V^+ e^{\gamma l} + \Gamma(l) V^+ e^{\gamma l}$$
$$V(l) = V^+ e^{\gamma l}(1 + \Gamma(l))$$
$$I(l) = \frac{V^+}{Z_0} e^{\gamma l}(1 - \Gamma(l))$$

$$I = I^+ e^{\gamma l} + I^- e^{-\gamma l}$$
$$= \frac{V^+}{Z_0} e^{\gamma l} - \frac{V^-}{Z_0} e^{-\gamma l}$$
$$= \frac{V^+}{Z_0} e^{\gamma l}\left(1 - \frac{V^-}{V^+} e^{-2\gamma l}\right)$$
→ derive:
$$= \frac{V^+}{Z_0} e^{\gamma l}(1 - \Gamma(l))$$

$$Z(l) = Z_0 \frac{(1 + \Gamma(l))}{(1 - \Gamma(l))}$$

find out Γ_L = in terms of Z_0.

At $l = 0$, $Z(l)$ must be Z_L

$$Z_L = Z_0 \frac{(1 + \Gamma_L)}{(1 - \Gamma_L)}$$

$$\boxed{\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

for no reflecⁿ: <7.8.24>

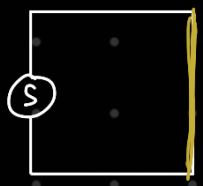
$Z_L = Z_0$ → w/I doesn't depend on l → interesting independency.?
↳ makes $Z_{input} = Z_0$

↓
Z_0 → depends on RLGC w/I is perm
↓
max power transfer & min reflecⁿ (signal integrity)

INTERPRETING Γ_L

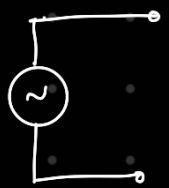
1. when $Z_L = Z_0$, $\Gamma_L = 0$; no reflection <matched impedance>
not all the power transfers from source to load due to attenuation (no resistor dissipation?)
↓

2. Short circuit-ed. (load is short-circuited)



$\Gamma_L = -1$; $Z_L \to 0$
no voltage delivered to load, All voltage reflected back.
1 → full reflecⁿ, (−) → 180° phase shift wrt incident voltage.
↓
<its $\Delta\phi$ not ϕ.>

3. Open circuited



$\Gamma_L = 1$; $Z_L \to \infty$
All voltage reflected back
+1 ⇒ no phase shift for reflected voltage w.r.t incident voltage.

$$V = V^+ e^{\gamma l} + V^- e^{-\gamma l}$$

$$= V^+ e^{\gamma l} + V^+ e^{\gamma l}\,\Gamma(l)$$

$$= V^+ e^{\gamma l}\,(1+\Gamma(l))$$

$$V(l) = V^+ e^{\gamma l}\,(1+\Gamma_L e^{-2\gamma l})$$

$$I(l) = \frac{V^+ e^{\gamma l}}{Z_0}\,(1-\Gamma_L e^{-2\gamma l})$$

$$Z(l) = \frac{V(l)}{I(l)} = Z_0\,\frac{(1+\Gamma_L e^{-2\gamma l})}{(1-\Gamma_L e^{-2\gamma l})}$$

$$\Gamma(l) = \frac{V^-}{V^+} e^{-2\gamma l} = \Gamma_L e^{-2\gamma l}$$

Always load your circuit b4 you turn on the source.

↓

o/w the source gets damaged due to reflecn.

$$= Z_0\left(\frac{1+\frac{Z_L-Z_0}{Z_L+Z_0}e^{-2\gamma l}}{1-\frac{Z_L-Z_0}{Z_L+Z_0}e^{-2\gamma l}}\right)$$

$$\cosh(x) = \frac{e^x+e^{-x}}{2}$$

$$\sinh(x) = \frac{e^x-e^{-x}}{2}$$

$$\cosh^2(x) - \sinh^2(x) = 1$$

$$= Z_0\left[\frac{e^{+\gamma l}(Z_L+Z_0) + e^{-\gamma l}(Z_L-Z_0)}{(Z_L+Z_0)e^{\gamma l} - (Z_L-Z_0)e^{-\gamma l}}\right]$$

$$= Z_0\left[\frac{Z_L(e^{\gamma l}+e^{-\gamma l}) + Z_0(e^{\gamma l}-e^{-\gamma l})}{Z_L(e^{\gamma l}-e^{-\gamma l}) + Z_0(e^{\gamma l}+e^{-\gamma l})}\right]$$

Lossless

$\alpha = 0$

$\beta = \omega\sqrt{LC}$

↓

cosh ⟶ cos

↓

when argument becomes purely imaginary

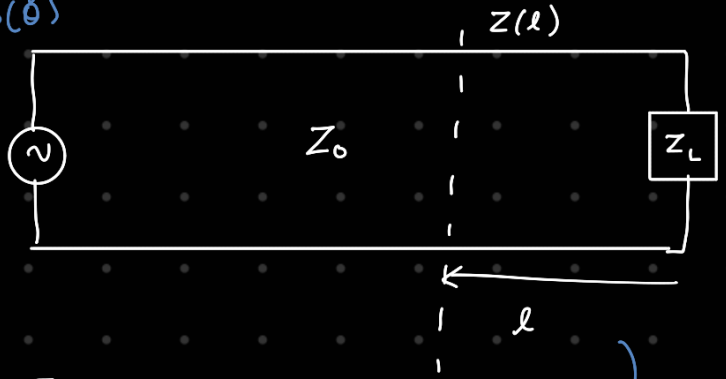
$\cosh(j\theta) = \cos(\theta)$

$$= Z_0\left[\frac{Z_L\cosh(\gamma l) + Z_0\sinh(\gamma l)}{Z_L\sinh(\gamma l) + Z_0\cosh(\gamma l)}\right]$$

$$= Z_0\left[\frac{Z_L + Z_0\tanh(\gamma l)}{Z_L\tanh(\gamma l) + Z_0}\right]$$

= in terms of one variable (l)

Learn how to do cosh tanh for complex nos. in calc



for $Z_L = Z_0$

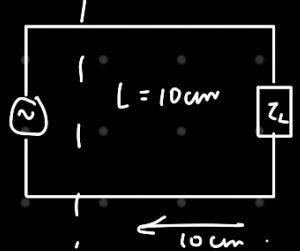
$\Rightarrow Z_L = Z(l) = Z_0$

→ idea of Impedance transformation: The length l of the line tranforms the impedance from Z_L to $Z(l)$

$\alpha \neq 0$, $\Gamma_L = 0$ → matched impedance

lossless lines? → $\alpha = 0$

Problem:



$Z_L = 100 + j100\ \Omega$

$Z_0 = 50\ \Omega$

. find Z at $l = +10$ cm.

we are finding transformed impedance at the locan of the source.

- R, L, G, C we used last week.

$w =$

$1GHz = v$

$\lambda = \frac{c}{v} = \frac{3\times10^8}{10^9} = 0.3m = 30cm.$

check Q. what?

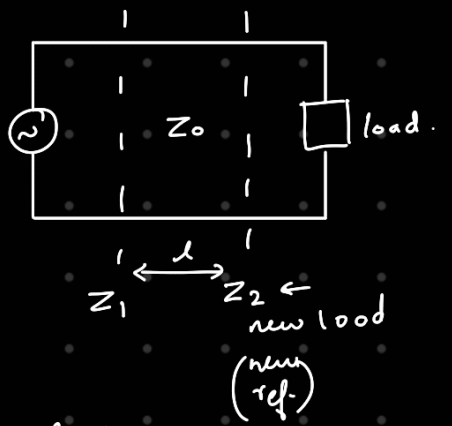
$f = 10GHz, \quad \lambda = 3cm$

$l = 100cm,$

$f_{given} = 10GHz$ }

NORMALIZED IMPEDANCE:

Given Z_2 at any arbitary locan we can find Z_1 at any other locan.



Locan of load is the reference $l = 0$

$$Z_1 = Z_0\left[\frac{Z_2 + Z_0 \tanh \gamma l}{Z_2 \tanh \gamma l + Z_0}\right]$$

$$Z_2 = Z_0\left[\frac{Z_1 + Z_0 \tanh(-\gamma l)}{Z_0 + Z_1 \tanh(-\gamma l)}\right]$$

$$\bar{Z}_1 = \frac{Z_1}{Z_0}$$

$$\bar{Z}_2 = \frac{Z_2}{Z_0}$$

$\alpha \rightarrow$ measure of signal loss per unit length due to R and G of the line.

Why matched impedance is not lossless lines?

(1) matched impedance → ensure no reflecn but not for losses w/n the Tx line itself.
↓
if $R \neq 0$, $G \neq 0$ there will be power loss due to them.

lossless lines → has no internal power loss.

and matrix inversion

DO THAT Q & CALC

1. solve complex
2. solve tanh, cosh, sinh.

Low-Loss & Loss-less – Standing wave properties

↳ $R=0,\ G=0,\ \gamma = j\beta = j\omega\sqrt{LC}$

$Z_0 = \sqrt{\frac{L}{C}}$ $\quad \alpha = 0$

$G \ll \omega C$
$R \ll \omega L$

$$\gamma = \alpha + j\beta = \sqrt{(R+j\omega L)(G+j\omega C)}$$

$$= \sqrt{j\omega L\left(1-j\frac{R}{\omega L}\right) j\omega C\left(1-j\frac{G}{\omega C}\right)}$$

$$\sim j\omega\sqrt{LC}\left(1-j\frac{R}{2\omega L}\right)\left(1-j\frac{G}{2\omega C}\right)$$ Taylor expansion

$$\sim j\omega\sqrt{LC}\left(1-j\frac{R}{2\omega L}-j\frac{G}{2\omega C}\right)$$

$$\sim \omega\sqrt{LC}\left(j+\frac{R}{2\omega L}+\frac{G}{2\omega C}\right)$$

$$\beta \sim \omega\sqrt{LC} \qquad \alpha = \frac{R}{2}\sqrt{\frac{C}{L}} + \frac{G}{2}\sqrt{\frac{L}{C}}$$

Low loss line has same β as loss-less line.
Usually, $\alpha \ll \beta$

α: unit → Neper/m.
Decibel unit: convert from Neper to dB

$\alpha = 1$ Neper/m (let)

$\alpha_{dB} = -20\log(e^{-1}) = 8.68$

$V_{out} = V_{in} e^{-\alpha l}$.

$V_{out} = V_{in} e^{-1}$

1 Neper/m = 8.68 dB/m.

Voltage loss:

dB loss in vol. : $-20\log_{10}\frac{V_{out}}{V_{in}}$ ↑ to get a +ve qun.

$\therefore V_{out} < V_{in}$.

$\alpha_{Np} = \ln\left(\frac{\text{Initial amp}}{\text{final amp}}\right) = e^{\alpha_{Np}} = \left(\frac{\text{Ini.}}{\text{final}}\right)$

$\alpha_{dB} = -20\log_{10}(e^{-1})$?

$\alpha_{Db} = 20\log_{10}\left(\frac{\text{Initial amp}}{\text{final amp}}\right) = 20\log\left(e^{\alpha_{Np}}\right)$

Power loss

$\alpha_{dB} = -10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$

$\alpha_{Db} = 20\,\alpha_{N_{Np}}\log e$

Problem: for voltage loss → $\alpha_{Db/m} = 8.68 \times \alpha_{Np/m}$ → for power loss $\alpha_{Db/m} = 4.34\,\alpha_{Np/m}$

$R = 0.2\ \Omega/m$
$L = 0.3\ \mu H/m$
$G = 0$
$C = 15\ pF/m$

$f = 10$ GHz
$Z_L = 100 + j100\ \Omega$

$l = 100$ cm.

$Z_0 = \sqrt{\frac{R+j\omega L}{j\omega C}}$

Transformed Z at 100 cm, 10 cm, 3 cm.

(1) $Z_0 = \sqrt{\frac{R+j\omega L}{G+j\omega C}}$



$\omega = 2\pi \times 10^{10} \frac{rad}{s}$.

$\omega L = 2\pi \times 10^{10} \frac{rad}{s} \times 0.3\ 10^{-6} \frac{H}{m}$.

$= 0.6\pi \times 10^4 \frac{\Omega}{m}$

$\omega C = 2\pi \times 10^{10} \frac{rad}{s} \times 15 \times 10^{-12} \frac{F}{m}$.

$= 0.3\pi\ \Omega^{-1}/m$.

$Z_0 = \sqrt{\frac{0.2 + j\ 0.6\pi \times 10^4}{j\ 0.3\pi}} \sim \sqrt{2 \times 10^4} = \sqrt{2} \times 10^2\ \Omega/m$.

$= \sqrt{\frac{1.9 \times 10^4}{9.4 \times 10^{-1}}}$ $\quad -8.7 \times 10^{-1}$

$= 2 \times 10^4$ $\quad -8.8 \times 10^{-1}$

(1) $r_1 \angle \theta_1 = 1.88 \times 10^4 \angle 90°$

(2) $r_2 \angle \theta_2 = 0.94 \angle 90°$

(3) $\frac{r_1}{r_2} \angle \theta_1 - \theta_2 = 2 \times 10^4 \angle 0°$

(4) $\text{rect}\left(\frac{r_1}{r_2} \angle \theta_1 - \theta_2\right)$

(1) $r_1 \angle \theta_1 =$

(2) $r_2 \angle \theta_2 =$

(3) $\frac{r_1}{r_2} \angle \theta_1 - \theta_2 =$

(4) $\text{rect}\left(\frac{r_1}{r_2} \angle \theta_1 - \theta_2\right)$

$= \sqrt{\frac{r_1 \angle \theta_1}{r_2 \angle \theta_2}} = 141.42\ \Omega$

$Z_0 = 141.42\ \Omega$ $\quad Z(l)$

(2) Calculate α, β

$\gamma = \sqrt{(R+j\omega L)(G+j\omega C)}$

$= \sqrt{(0.2 + j\ 0.6\pi \times 10^4)(0 + j\ 30\pi \times 10^{-2})}$

$= \sqrt{-1.77 \times 10^4 + j\ 0.1884}$.

$= \sqrt{1.77 \times 10^4 \angle 179.9994}$

$$= \quad 6.966 \times 10^{-4} + j\, 1.33 \times 10^{2}$$

$$\alpha = 6.99 \times 10^{-4}$$

$$\beta = 1.33 \times 10^{2}$$

$$Z(l_m) = Z_0 \left(\frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right)$$

$$= Z_0 \left(Z_L + 141.200 \angle 1.27^{-4} \right)$$

A guide to using the calculator: $x+iy$

POL $(x, y) = |x+iy|$

↳ alpha F → $\angle(x+iy)$

for sqrt,

rect $\left(\sqrt{|x+iy|}, \frac{\angle(x+iy)}{2}\right)$

$= |\sqrt{x+iy}|$

↳ alpha, F → $\angle\sqrt{x+iy}$

STANDING WAVE PROPERTIES

<9.8.24>

Low-loss lines

$$V(l) = V^+ e^{j\beta l} \left(1 + \Gamma_L e^{-j2\beta l}\right)$$

$$I(l) = \frac{V^+}{Z_0} e^{j\beta l} \left(1 - \Gamma_L e^{-j2\beta l}\right)$$

$$\Gamma(l) = Z_0 \left(\frac{Z_L + Z_0}{Z_L - Z_0} \right)$$

$$\Gamma(l) = \Gamma_L e^{-j2\beta l}$$



Γ_L is in general complex.

$$\Gamma_L = |\Gamma_L| e^{j\phi_L}$$

$$V(l) = V^+ e^{j\beta l} \left(1 + |\Gamma_L| e^{j(\phi_L - 2\beta l)}\right)$$

$\Gamma_L = 0$ (no reflecn)

↓

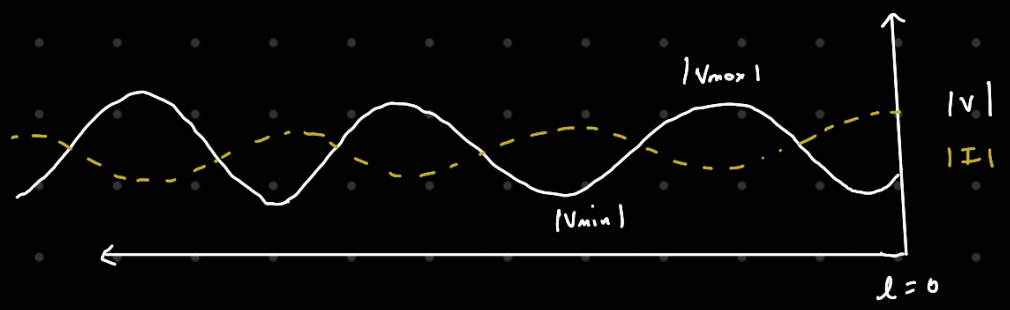
when this is 1 and it is out of phase

↓

min Amp = 0

$$|V_{max}| = |V^+| \left(1 + |\Gamma_L|\right)$$

$$|V_{min}| = |V^+| \left(1 - |\Gamma_L|\right)$$



microphones

↓

capacitive / inductive

electric signals → sound

↓

mostly inductive.

↓

eg: Headphones Speakers.

PCBs → Printed circuit Boards

↓

primarily resistive.

L + C

↳ oscillators.

why are they out of phase by 90°

↝ ?

$V_{max} \rightarrow I_{min}$.

$V_{min} \rightarrow I_{max}$

↓

in the case of standing wave.

↳ $|\Gamma_L| = 1$?

Ratio, $\frac{|V|_{max}}{|V|_{min}} \equiv S$, (Voltage Standing Wave Relationship)

Ideally $S = 1$? $= \frac{1+|\Gamma_L|}{1-|\Gamma_L|} \Rightarrow$ no reflecn $\rightarrow$ no max / min $\rightarrow$ no standing wave.

↓ perfect impedance match

$\Gamma_L = 0$

when impedance match / No reflection,

$S = 1$

o/w $S > 1$

for full reflecn

$\Gamma_L = 1$

$S = \infty$

open circuit $S = \infty$; $\Gamma_L = -1$ } complete reflecn

short circuit $S = \infty$; $\Gamma_L = 1$ }

better

$S = 2 > 100$?

because it indicates a better impedance match.

XII std

$V_{rms} I_{rms} \cos\phi$.

POWER DELIVERED TO THE LOAD.

$$P = \frac{1}{2} Re\{V I^*\}$$

$$= \frac{1}{2} Re\{V(l)\, I^*(l)\}$$

$$= \frac{1}{2} Re\left\{ V^+ \left(1 + |\Gamma_L| e^{j\phi_L}\right) \frac{V^+}{Z_0} \left(1 - |\Gamma_L| e^{-j\phi_L}\right) \right\}$$

$$= \frac{|V^+|^2}{2Z_0} Re\{ 1 - |\Gamma_L| e^{-j\phi_L} + |\Gamma_L| e^{j\phi_L} - |\Gamma_L|^2 \}$$

$$= \frac{|V^+|^2}{2Z_0} \{ 1 - |\Gamma_L| \cos(\phi_L) + |\Gamma_L| \cos(\phi_L) - |\Gamma_L|^2 \}$$

$$P_L = \frac{|V^+|^2}{2Z_0} (1 - |\Gamma_L|^2)$$ → open / short $|\Gamma_L| = 1$ no power delivered.

max power to load when,

$\Gamma_L = 0 \rightarrow$ no reflecn

MAXIMUM POWER TRANSFER THEOREM:

★ For a complex load, the load must be complex conjugate of source

let $Z_S = R_S + jX_S$

$Z_L = R_L + jX_L$

$$P = \frac{|V_S|^2}{|Z_S + Z_L|^2} R_L$$

making power factor 1.

↑

(eliminating reactive power)

↓

$R_S = R_L$ and $X_S = -X_L$

$\therefore Z_S = \overline{Z_L}$

minimises $|Z_S + Z_L|^2$ ←

for a complex load, the load must be complex conjugate of source?

why, some phase Δ.

↓

applicable only when Z_L is transformed

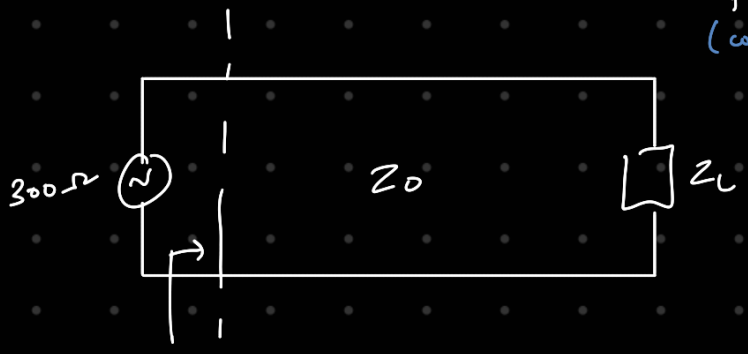
↳ For Tx lines this works b/w Z_0 and Z_L transformed to a point near the source

i.e., $Z_0 = \overline{Z}_{L\,Transformed}$

to near the source & then apply the $Z_0 = \overline{Z}_{L\,Tran}$.

very difficult to make Z_0 to Z_L to $Z_{free\ space}$ (communicate from source to load to free space)

$\sqrt{\frac{E}{H}} = Z_{free\ space}$.



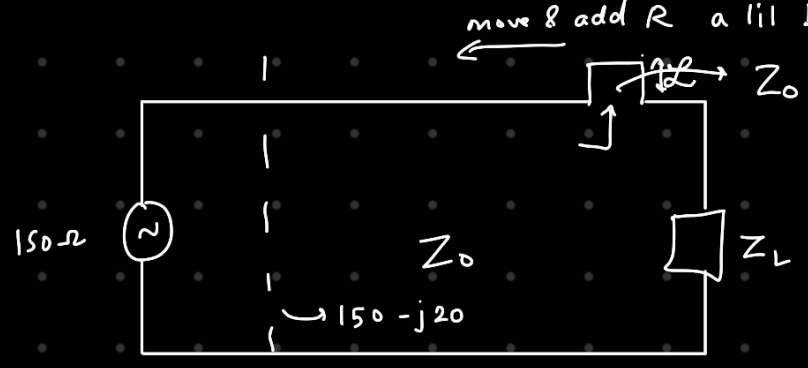
$150\Omega + j20$ → you would connect $150\Omega - j20$.

↓

but cannot put bulk resistance for high frequency circuits.

↓

you Δ the design



STUB MATCH

$\alpha = 0$ → lossless assump^n?

$$Z(\ell) = Z_0 \left[\frac{Z_L \cos\beta\ell + j Z_0 \sin\beta\ell}{Z_0 \cos\beta\ell + j Z_L \sin\beta\ell} \right]$$ for SC. $Z_L = 0$

$Z_{sc} = j Z_0 \tan\beta\ell$ → choose ℓ ST it cancles out w/ $j20$

Stub → a segment of Tx line that can create a purely reactive impedance depending on its length (ℓ) and whether it is open circuit / short circuit.

↓

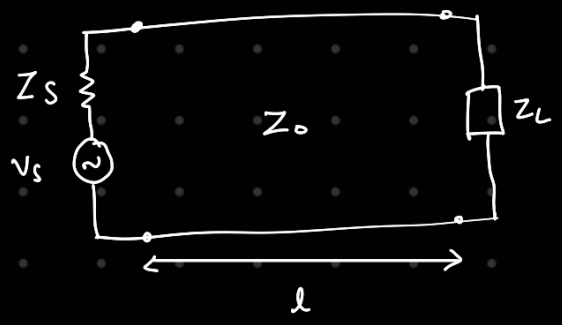
distance chosen from the load is ST the Z at that point on the Tx line can be matched to Z_0 of the line when the stub is added in ||el.

↓

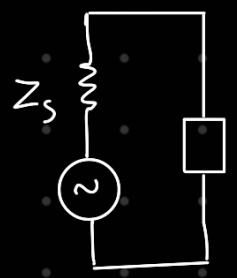
It effectively makes the total impedance seen by the source = Z_0. → w/I minimizes reflections, and maximises power transfer.

<12·8·24>

next week tuesday ↓ finish this weekend.



≡



→ Look-in impedance AKA Transformed impedance.

H.W

find P along the line.

✓ resistive ← prove $Re\{\ \} = k$

✓ imaginary ← reactive power = keeps oscillating. $[V(l)\, I^*(l)]$

stored power.

$$Z_L' = Z_0 \left[\frac{Z_L \cos\beta l + j Z_0 \sin\beta l}{Z_0 \cos\beta l + j Z_L \sin\beta l} \right]$$

$$V(l) = V^+ e^{j\beta l} \left[1 + \Gamma_L e^{-j2\beta l} \right]$$

$$I(l) = \frac{V^+}{Z_0} e^{j\beta l} \left[1 - \Gamma_L e^{-j2\beta l} \right]$$

$$\frac{V(l)}{I(l)} = Z(l) = Z_0 \left[\frac{1 + \Gamma_L e^{-j2\beta l}}{1 - \Gamma_L e^{-j2\beta l}} \right]$$

$$= Z_0 \left[\frac{(Z_L + Z_0) e^{j\beta l} + (Z_L - Z_0) e^{-j\beta l}}{(Z_L + Z_0) e^{j\beta l} - (Z_L - Z_0) e^{-j\beta l}} \right]$$

$$= Z_0 \left[\frac{Z_L (e^{j\beta l} + e^{-j\beta l}) + Z_0 (e^{j\beta l} - e^{-j\beta l})}{Z_L (e^{j\beta l} - e^{-j\beta l}) + Z_0 (e^{j\beta l} + e^{j\beta l})} \right]$$

$$\tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$= Z_0 \left[\frac{Z_L \cos\beta l + j Z_0 \sin\beta l}{Z_0 \cos\beta l + j Z_L \sin\beta l} \right]$$

if $x = iy$:
$\sinh(iy) = i\sin(y)$
$\cosh(iy) = \cos(y)$
$\tanh(iy) = i\tan(y)$

$$= Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right]$$

tanh → tan as $\alpha = 0$

(1) $Z(l + \lambda/2) = Z(l)$ ⇒ sufficient to understand the behaviour of $\lambda/2$ section of the line.



$\cos(2kx)$ ← term in standing wave.

$2\beta l = 2\pi \cdot m$

$2 \cdot \frac{2\pi}{\lambda} \cdot l = 2\pi$

$l = \frac{\lambda}{2}$ ← that func. is periodic w/ $\lambda/2$

(2) Normalized impedance: $\frac{Z(l)}{Z_0} = \bar{Z}$

↓ inverts every $\lambda/4$?

↓ (look at the periodicity)

Z_0 is real → B/C lossless

(3) $|Z(l)|_{max} = \frac{V_{max}}{I_{min}} = \frac{|V^+|(1 + |\Gamma_L|)}{\frac{|V^+|}{|Z_0|}(1 - |\Gamma_L|)}$

$= |Z_0| \left(\frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} \right) = Z_0 S.$

↗ $Z(l)$ could be complex?

$|Z(l)|_{min} = \frac{Z_0}{S}$

This is important to see where to cut the tx line to do impedance matching.

You can use this to find unknown impedance.

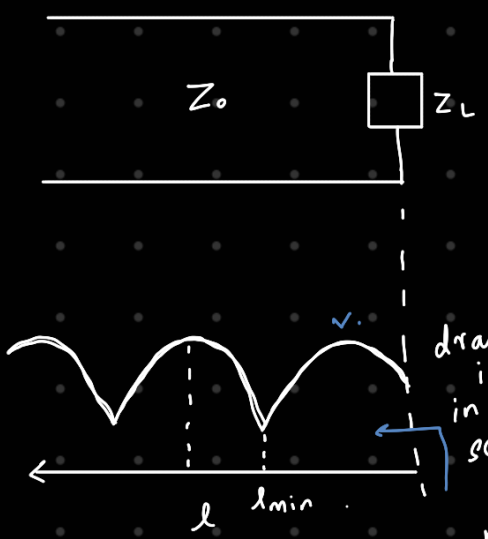
why 77Ω?

BNC cable

SMA cable.

↑ RF cables.

↓ a bench in Tx line.



drawn it in logarithmic scale

finding min is easier than finding a max (?)

↓ min is sharp.

find log amp since find min sharper.

1. Connect unknown impedance to a Tx line of known Z_0
2. Locate the position of first (voltage) minimum.

3. measure $|V|_{max}$, $|V|_{min}$.

$$S = \frac{|V|_{max}}{|V|_{min}}$$

$$\therefore R_{min} = \frac{|Z_0|}{S}$$

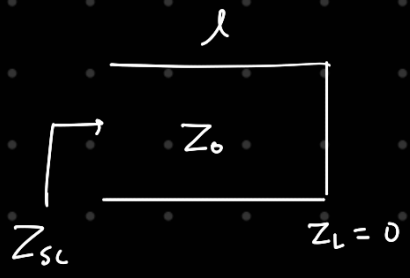
↳ B/C Z_0 is purely real for a loss-less line → ∴ no need of ||

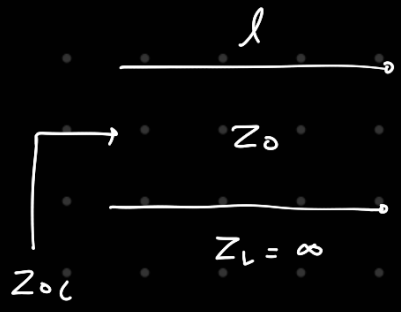
(why don't we know the reactance?)

R_{min} must be the look-in or transformed impedance at l_{min}.

$$Z_L = Z_0\left[\frac{\frac{Z_0}{S} + j Z_0 \tan(-\beta l_{min})}{Z_0 + j\frac{Z_0}{S}\tan(-\beta l_{min})}\right]$$

4.

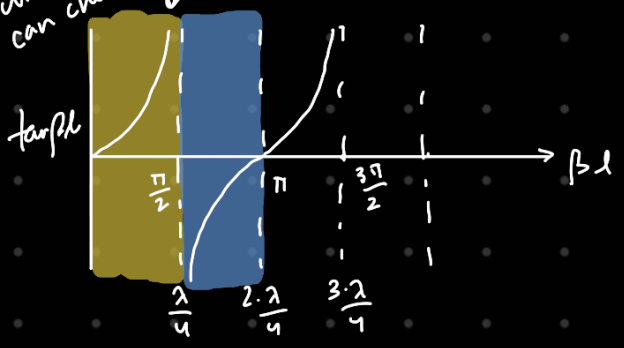




$$Z_{sc} = j Z_0 \tan\beta l$$

this is the fundamental character i.e. we can choose harmonics!

purely reactive.



$$\beta l = \frac{\pi}{2}$$

$$\frac{2\pi}{\lambda}\cdot l = \frac{\pi}{2}$$

$$l = \lambda/4.$$

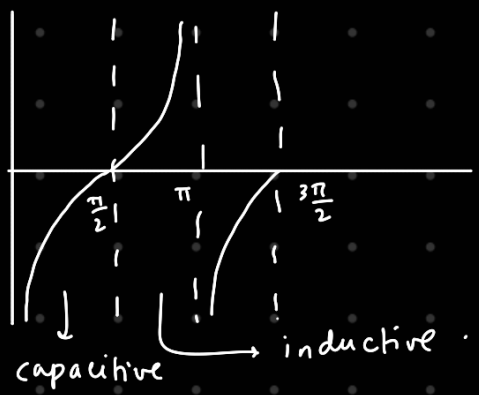
$$0 < l < \lambda/4$$

↓

Short circuit → inductor (behaves like) (+ve reactance)

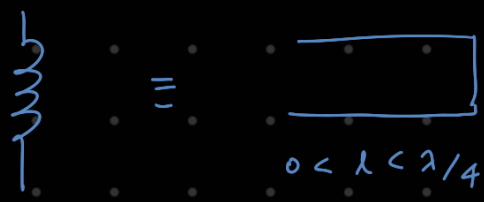
$$Z_{oc} = j\frac{Z_0}{j^2}\cot\beta l$$

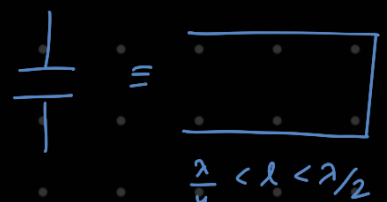
$$= -jZ_0\cot\beta l.$$



Inductors & capacitors can be built/designed w/ SC/OC Tx lines.

$$\frac{\lambda}{4} < l < \frac{\lambda}{2}$$ → short circuit → capacitor (-ve reactance)

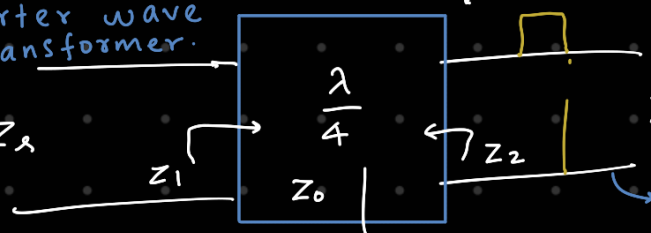




If I can make L & C? I can make a resonant ckt.

Resonant circuits can be built w/ Tx lines as circuit elements.

5. Quarter wave Transformer.



put Tx line, what line do I use ST look-in impedance

after every 2π phase.

$\frac{\lambda}{4}$ → repeating every $\frac{3\lambda}{4}$.

these aren't TL

I'll use odd multiples of $\lambda/4$

Z_s, Z_L are resistive.

$$\beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4} = \frac{\pi}{2}$$

$$\sin\beta l = 1$$
$$\cos\beta l = 0.$$

period of this is $\lambda/2$?

$$Z_1 = Z_0 \left[\frac{Z_L \cos\beta l^{\circ} + j Z_0 \sin\beta l}{Z_0 \cos\beta l^{\circ} + j Z_L \sin\beta l} \right]$$

$$Z_1 = \frac{Z_0^2}{Z_L}$$

↑ transformed from Z_L thru $\lambda/4$.

for impedance matching, (source side)

$$Z_s = Z_1$$

$$\Rightarrow \quad Z_s = \frac{Z_0^2}{Z_L}$$

$$Z_0 = \sqrt{Z_s Z_L}$$ → if choose this Z_s is matched.

$$Z_2 = \frac{Z_0^2}{Z_s}$$

for matching. (load side)

$$Z_2 = Z_L$$

$$\Rightarrow \quad Z_L \cdot Z_s = Z_0^2$$

$$Z_0 = \sqrt{Z_s Z_L}$$

for reactive impedance, we need conjugate to match.

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(ix) = \cos x$$
$$\sin(ix) = i\sin(x)$$

$$l < 13.9 - j12.1\ \Omega$$

even w/o matching if $\lambda/4$

$Z_0 = 50\Omega$

$Z_L = 100\Omega$

$Z_{look\ in} = 25\Omega$

Quarterwave transformer: (when load is fixed, you design the line) <14.8.24>

introduce a TL line $l = \frac{\lambda}{4}$ or $(2n+1)\frac{\lambda}{4}$ w/ $n = 0, 1, 2, 3, \ldots$ B/w Z_S & Z_L

$$Z_0 = \sqrt{Z_S Z_L}$$

VNA vector network analysis.

Halfwave Transformer

$$\beta l = \frac{2\pi}{\lambda} \cdot \frac{\lambda}{4}$$

$Z_L \longrightarrow$ transform to point near source also gives Z_L as length is $\lambda/2$ B/w the 2 pts.

↓ and for z matching $Z_S = Z_L$

$Z_0 = \overline{Z_L}$ ↓ only for power. ↓ when it comes to reflecn, this is not how it works.

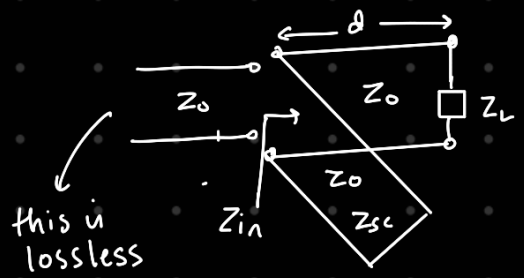
tanh ↕ tan.

not a strict rule whether open / SC is favoured.

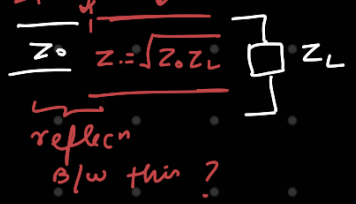
$Z_1 = Z_L$

$Z_2 = Z_S$

IMPEDANCE MATCHING REACTIVE LOADS



Complex $R_L + jX_L$ ↓ can +ve/-ve



$Z_0 = \mathrm{Re}\{Z_{in}\}$

$Z_{in} = Z_{sc}$. (Stub $\rightarrow \parallel^{el}$ OR series)

For impedance matching:

$$Z_{in} = Z_0 \left[\frac{Z_L \cos\beta d + j Z_0 \sin\beta d}{Z_0 \cos\beta d + j Z_L \sin\beta d} \right]$$

having Z_0 comp ↓ makes it not loss less ↓ what about Z_L being complex.

$Z_0 = Z_{in} \parallel Z_S$ you could choose open ckt line as well.

$$Y_0 Z_0 = Y_{in} Z_0 + Y_{sc} Z_0$$

$$1 = \underbrace{Z_0 Y_{in}}_{Z_0 \rightarrow \text{is real.}} + \underbrace{Y_{sc} Z_0}_{\rightarrow jX_{sc}}$$

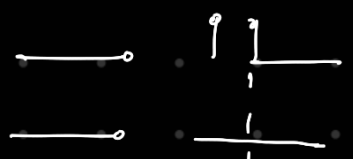
Complex $R_{in} + jX_{in}$

$\rightarrow jC$; is a func^n of l.

$$R_{in} + jX_{in} \equiv A + jB$$

is a func^n of l

Choose d ST $A=1$, find B
— " — l ST $C=-B$



↑ we don't know what is the value?

we won't have steady state?



switch on & off

acts like an echo

↓

this disrupts communica^{n} of other signals

Power along the line:

Prove $\quad \mathrm{Re}\{P\}$ = resistive = constant.

$\mathrm{Im}\{P\}$ = reactive = keeps oscillating.

$$P = \frac{1}{2}\left[V(l)\, I^{*}(l)\right]$$

$$V(l) = V^{+} e^{j\beta l}\left[1 + \Gamma_L e^{-j2\beta l}\right]$$

$$I(l) = \frac{V^{+}}{Z_0} e^{j\beta l}\left[1 - \Gamma_L e^{-j2\beta l}\right]$$

$$= \frac{1}{2} V^{+} e^{j\beta l}\left[1 + \Gamma_L e^{-j2\beta l}\right] \frac{V^{+}}{Z_0} e^{-j\beta l}\left[1 - \Gamma_L e^{j2\beta l}\right]$$

$$= \frac{1}{2}\frac{V^{+2}}{Z_0}\left[1 - \Gamma_L e^{j2\beta l} + \Gamma_L e^{-j2\beta l} - |\Gamma_L|^2\right]$$

$$= \frac{|V^{+}|^2}{2Z_0}\Gamma_L \underbrace{\left(e^{-j2\beta l} - e^{j2\beta l}\right)}_{c - s - (c+s)} + \frac{|V^{+}|^2}{2Z_0}\left(1 - |\Gamma_L|^2\right)$$

$$= \frac{|V^{+}|^2}{2Z_0}\Gamma_L (-2j)\sin(2\beta l) + \frac{|V^{+}|^2}{2Z_0}\left(1 - |\Gamma_L|^2\right)$$

$$\Gamma(l) = \frac{V^{-}}{V^{+}} e^{-j\beta l} e^{j\beta l}$$

$$\Gamma_L = \frac{V^{-}}{V^{+}}$$

imaginary part ↓ depends on l

real part ↓ constant along line.

$\frac{e^{j\theta} - e^{-j\theta}}{2j} = \sin\theta$

$\frac{e^{x} - e^{-x}}{2} = \sinh(x)$

↓ proof.

$x = iy$.

$\frac{e^{iy} - e^{-iy}}{2} = \sinh(iy) = j\sin(y)$

$\frac{e^{j\theta} + e^{-j\theta}}{2} = \cos(\theta)$

$\frac{e^{x} + e^{-x}}{2} = \cosh(x)$

$x = iy$.

$\cosh(iy) = \frac{e^{iy} + e^{-iy}}{2} = \cos(y)$

$Z(l + \lambda/2) = Z(l) \longrightarrow$ for standing waves. ?

↓

for every $\frac{\lambda}{4} \Rightarrow Z_s = \frac{Z_0^2}{Z_L}$

$\therefore \quad Z_0 = \sqrt{Z_s Z_L}$ → for impedance matching.

o/w for every $\frac{\lambda}{4} \Rightarrow \left(\frac{Z_s}{Z_0}\right) = \left(\frac{Z_0}{Z_L}\right)$ normal impedance inversion.

$\overline{Z_s} = \frac{1}{\overline{Z_L}}$

the normal impedances inverts after every $\frac{\lambda}{4}$ while repeating every $\lambda/2$

$Z(l)$
↓
low less

